

Eric Gu - Software Engineer

AUSTIN, TX

(479) 573-6378

gudmaneric@gmail.com

www.linkedin.com/in/etgu/

<https://ericsportfolio.vercel.app/>

Skills

Programming Languages: C++/C, Rust, Java, Python, JavaScript, Go, R, SQL, Bash/Shell, ARM

Operating & Build Systems: Linux, Android, Vega, QNX, Open-Embedded/Yocto, Make/CMake

Frameworks & Tools: GStreamer, OpenCV, V4L2, HAL3/Camera2 API, TensorFlow, AWS, GCP

Experience

February 2022 - PRESENT

Amazon, Austin, TX - Software Development Engineer II

- Designing and optimizing Camera HAL, IPC, and framework components across smart vehicles, wearables, Echo devices, Ring, AR/VR, VegaOS, and Android.
- Lead and provide across teams for the following infrastructure:
 - Amazon Vision Framework: powering a camera framework for VegaOS embedded devices, providing an AI/ML-powered vision cloud-based platform optimized for camera-based scene understanding.
 - Echo Camera Service Framework: Own full-stack upgrading of camera services for over 100 million Amazon's Android Echo devices, enabling features like video calling and image recognition for the Computer Vision and Communications teams.
 - Multimedia Recorder Framework: Optimized the Multimedia Recorder Framework for next-gen devices with hardware encoders, reducing memory usage by 616MiB and decreasing CPU usage by 43% to enhance video/audio capture capabilities.
- Leading code reviews and yearly strategic roadmaps for the 8-person Vega Camera framework team; mentoring 2 junior engineers as a tech lead, and strengthening team collaboration and code quality through mentorship and weekly office hours.
- Promoted Q2 2024
- Leveraging C++, Java, Python, and frameworks and tools including Android, Linux, QNX, Open-Embedded/Yocto, GStreamer, OpenCV, and CMake.

January 2021 - December 2021

Emory University, Atlanta, GA - Software Engineer

- Directed a long-term data analytics project to increase underrepresented student success in Computer Science, achieving a 75% increase in retention rates after "CS170".
- Analyzed admissions and student data at Emory University using Python (pandas, numpy), TensorFlow for machine learning, natural language processing, and SQL for databases.

May 2021 - August 2021

Gigasheet, Boston, MA - Software Engineer Intern

- Secured Return Offer.
- Engineered 30 end-to-end automated feature tests that resulted in a 95% reduction in development site crashes.
- Utilized Cypress for front-end testing, Docker for app management and deployment, ClickHouse for database management, and Atlassian Bitbucket/Jira for task management. JavaScript/React/HTML/CSS on the front-end and Go/SQL on the back-end for application development.

June 2020 - August 2020

Wolfe Research, New York City, NY - Quantitative Engineer Intern

- Secured Return Offer.
- Worked closely with the lead research analyst, 5 other analysts, and 2 technology teams to develop 2 stock selection and global macro models.
- Mapped an R library on Custom Factor Attribution to Pure Python, impacting 100+ clients.
- Engineered a securities universe of over 4000 securities using R Query.
- Used Python, R, Linux, and Bash/Shell based in AWS to predict returns/risks of major asset classes, sectors, and investment styles.

Education

August 2018 - December 2021

Emory University, Atlanta, GA - Bachelor of Arts, Computer Science

- GPA: 3.7/4.0

Selected Projects

- **FireOS Camera Performance & Memory Optimization:** Engineered memory management in FireOS cameras, reducing cache misses and CPU overhead, leading to a 3% energy savings across all camera devices, cutting annual power consumption by ~71 GWh, saving users \$10 M+ annually.
- **VegaOS Pre-CI:** Designed and deployed a pre-CI system using Gitfarm CI plugins, cutting post-merge defects by 62% and preventing costly regressions. Saved \$600K annually for my team and \$5 M+ annually org-wide.
- **Smart Vehicle Multimedia Optimization:** Optimized Multimedia Recorder App, GStreamer camera and audio plugins using hardware encoder configurations and zero-copy buffer wrapping approach to eliminate redundant memory operations.
- **DOIP (Device Over IP) – Remote Access & Infrastructure Scaling:** Developed a Rust-based device remote access system, reducing device downtime by 40%, which is saving the org. \$6M+ annually. Built the infrastructure with AWS CDK for real-time device tracking, leveraging USBIP for lightweight, scalable remote debugging.